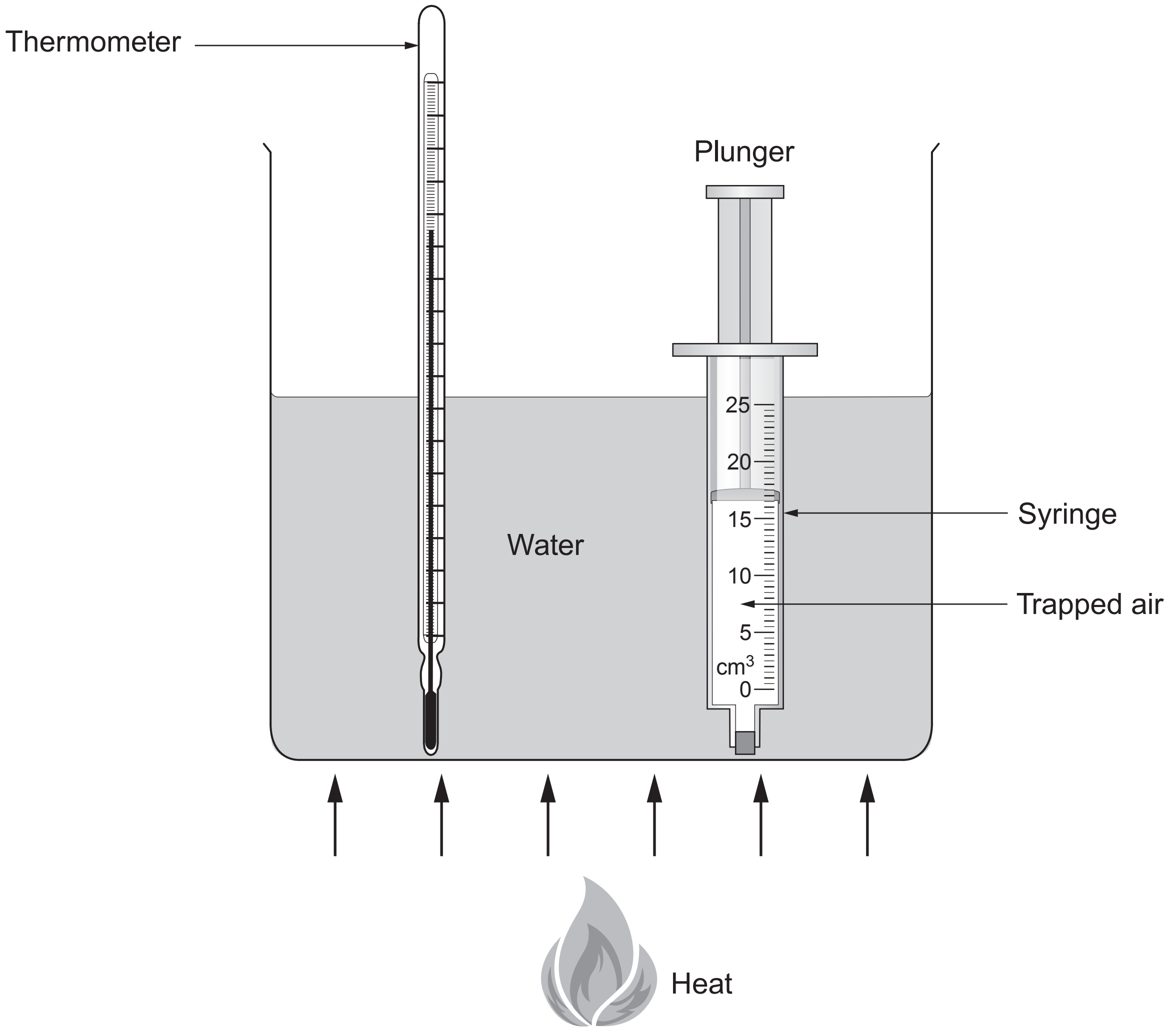
9. In a class experiment the volume of a fixed mass of air changes when it is heated.



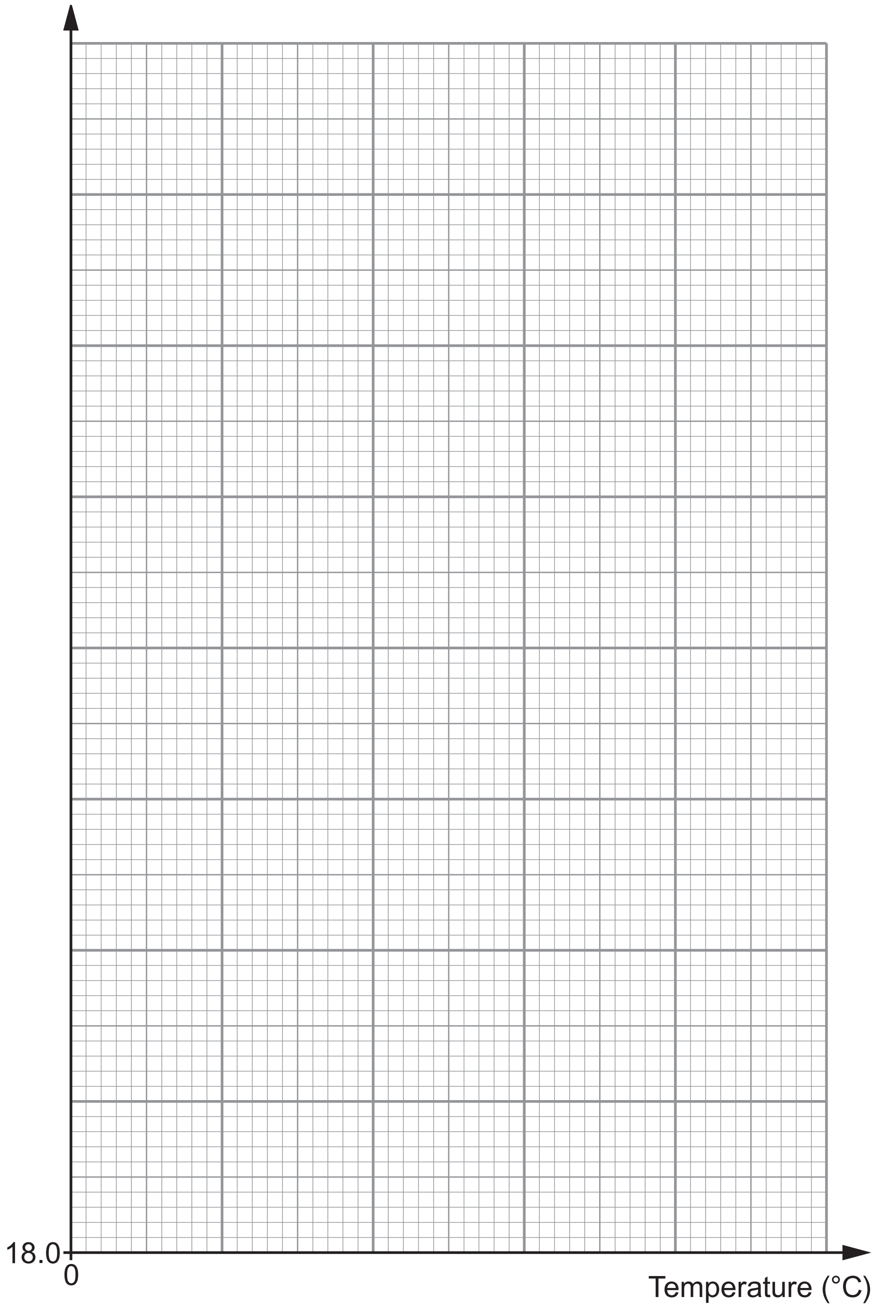
The air is trapped in a syringe which has been sealed at one end but the plunger at the other end is free to move in or out. The maximum volume the syringe can measure is 25.0 cm^3. The volume of trapped air in the syringe is measured as the temperature of the water is increased. The results are shown in the table below.

Temperature (°C)	Volume of trapped air (cm^3)
10	19.5
30	21.0
50	22.5
60	23.0
80	24.5

Examiner only

(a) Plot the data on the grid below and draw a suitable line.
(Note that the scale on the volume axis does not start at zero.) [3]

Volume of trapped air (cm^3)



(b) Explain, using your graph, whether the students can take a volume reading from the syringe at 100 °C. [2]

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QUESTION CONTINUES ON NEXT PAGE

 (3420UA0-1) **Turn over.**

Examiner only

(c) Explain how, if at all, the density of the trapped air changes as it is heated. [2]

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(d) Explain, in terms of the motion of molecules, why the volume of the trapped air increases as the temperature increases. [2]

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(e) The data in the table shows when the temperature increases from 10 °C to 60 °C the volume of trapped air increases from 19.5 cm^3 to 23.0 cm^3. Using only this information, calculate the temperature, in °C, at which the volume of the trapped air is zero. [3]

Temperature = ... °C

(f) State the name given to the temperature when the volume is zero. [1]

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END OF PAPER

